Database Systems

(Online Retail Store)

1.Introduction

Online retail stores deal with many connected types of data every day, such as customer accounts, product listings, categories, orders, payments, inventory, shipping information, and loyalty points. In this project, we selected the Online Retail Store domain to model how an ecommerce system stores and manages these records in an organized and accurate way.

The main goal of our project is to design and implement a database that supports the store's core operations, including managing customer profiles and delivery addresses, tracking orders and their detailed items, updating inventory quantities, recording payments, and storing loyalty point transactions. Our database design focuses on keeping the data consistent and reliable by using correct relationships between entities and enforcing integrity constraints, so the system avoids common issues such as invalid category references, missing customer links, or inconsistent transaction records

2. Conceptual Model Design (EER Diagram)

Entity and Attributes:

Customer	**(**CustomerID (PK), FirstName (VARCHAR), LastName(VARCHAR), Email (VARCHAR) , PhoneNumber(VARCHAR), Address(VARCHAR), Country(VARCHAR), LoyaltyPointsBalance (NUMBER))
Category	(CategoryID *(*PK)(NUMBER), CategoryName(VARCHAR))
Product	(ProductID *(*PK)(NUMBER), ProductName(VARCHAR), Description, Price(NUMBER), StockQuantity(NUMBER), CategoryID (FK)(NUMBER))
Orders	(OrderID (PK)(NUMBER), OrderDate(DATE), CustomerID (FK)(NUMBER), PaymentID (FK)(NUMBER), OrderStatus(VARCHAR))
Order_Detail	(OrderDetailID (PK)(NUMBER), OrderID (FK)(NUMBER), ProductID (FK)(NUMBER), Quantity(NUMBER), Price(NUMBER), Subtotal(NUMBER))
Inventory	(InventoryID (PK)(NUMBER), ProductID (FK)(NUMBER), QuantityAvailable(NUMBER), ReorderLevel(NUMBER))
Payment	(PaymentID (PK)(NUMBER), PaymentDate(DATE), PaymentMethod(VARCHAR), Amount(NUMBER), OrderID (FK)(NUMBER), PaymentStatus(VARCHAR))
Loyalty_Point	(LoyaltyTransactionID (PK)(NUMBER), Customer_ID (FK)(NUMBER), PointsEarned(NUMBER), PointsRedeemed(NUMBER), PointsDate (DATE))

Relationships:

1- Customer make order (1:N)
2- Order has OrderDetail (1:N)
3- Customer earns loyalityPoints (1:N)
4- Product belong to category (N:1)
5- Product has inventory (1:1)
6- Order has payment (1:1)
7- Order contain product (M:N)
8- 8- Order generate loyality points (1:N)

Specialization :

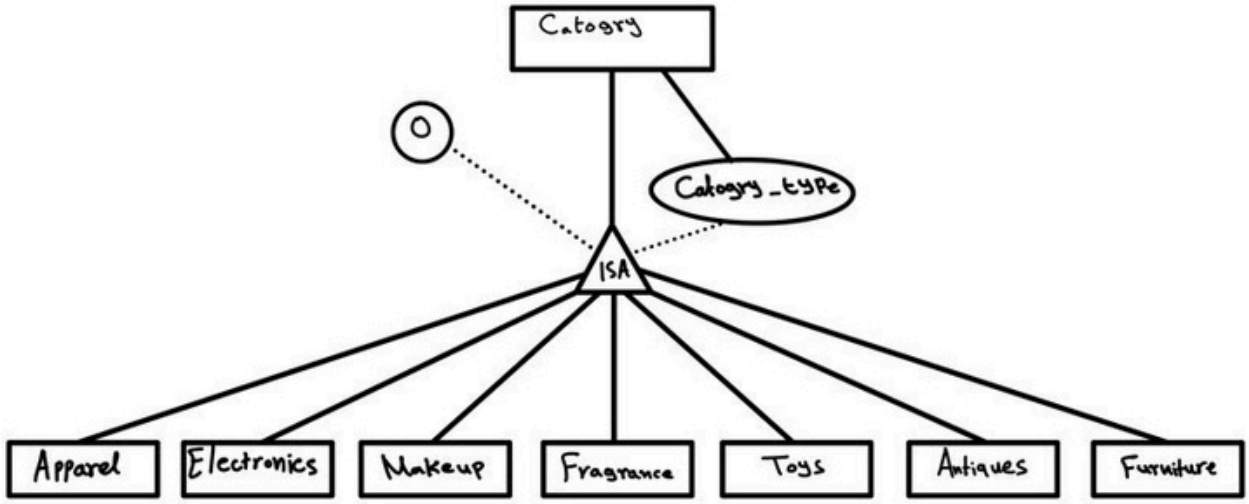
1-Category (Superclass) (Overlapping)
(Subclasses)
Apparel, Electronics, Makeup, Fragrances, Toys, Antiques and Furniture.

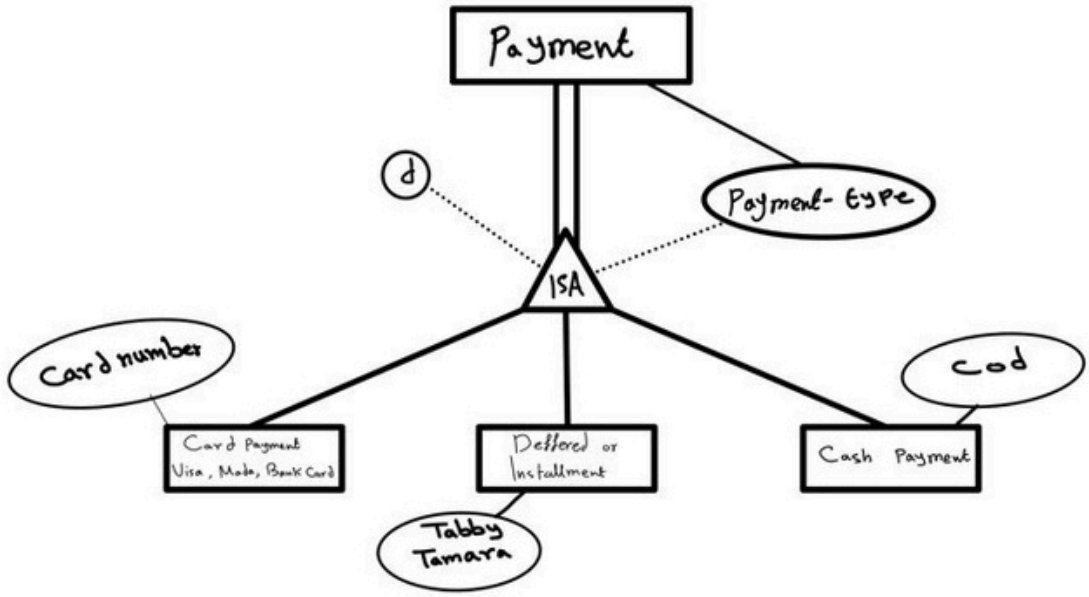
2-Payment (Superclass) (Disjoint)

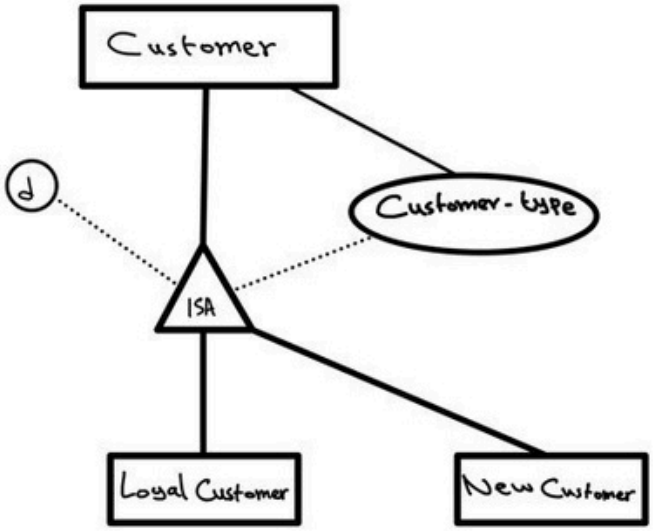
(Subclasses)
Card Payment (Visa\ Mada\Bank Card)(Card Number)
Deferred / Installment Payment (Tabby\Tamara)
Cash Payment (Cash on Delivery (COD))

3-Customer (Superclass) (Disjoint)
(Subclasses)
Loyal Customer, New Customer

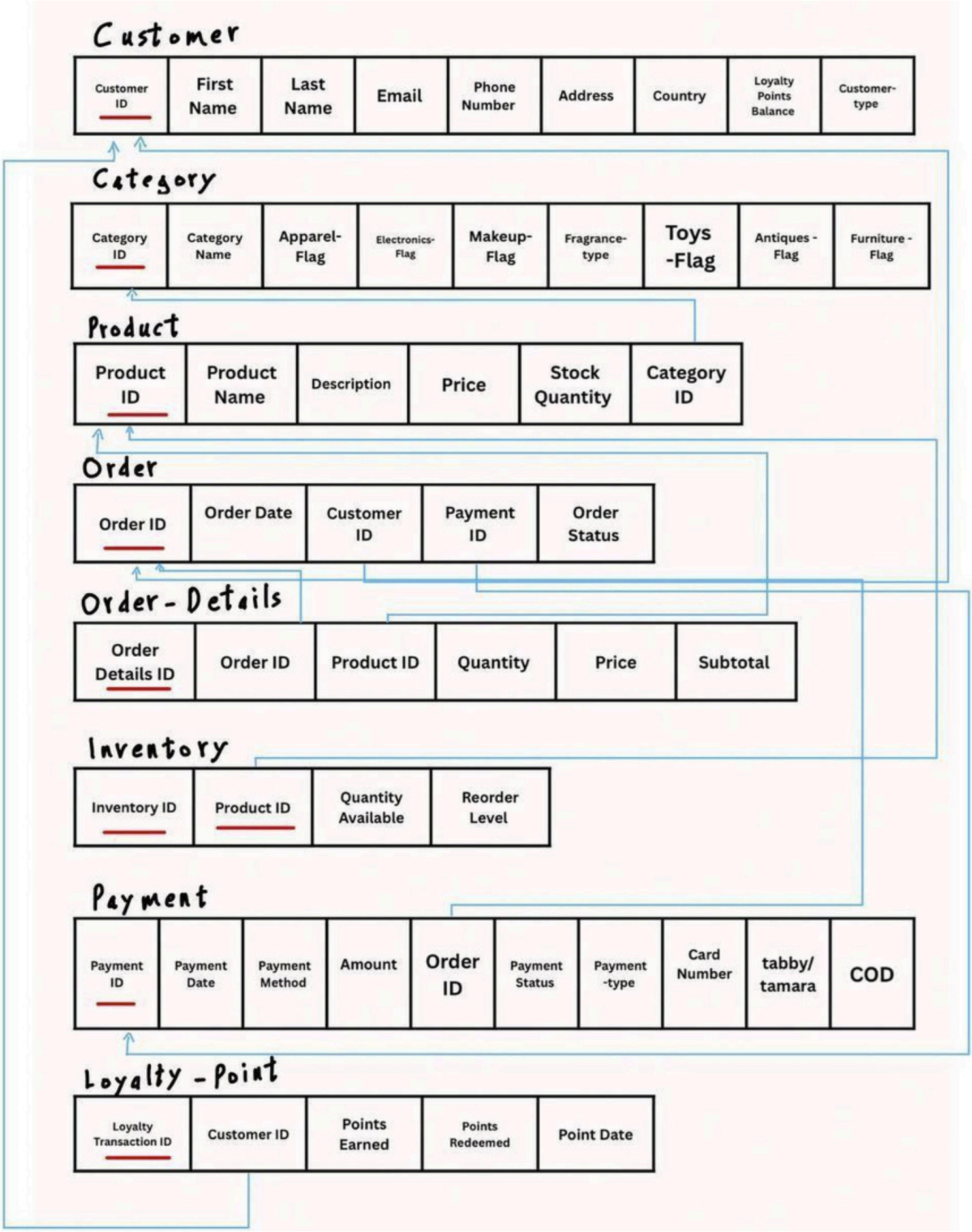
E-Commerce ER Diagram (Chen-style)


Catogry
O
Catogry_type
ISA
Apparel
Electronics
Makeup
Fragrance
Toys
Antiques
Furniture


Payment
d
Payment-type
ISA
Card number
cod
Card Payment
Visa, Mada, Bank Card
Deffered or
Installment
Cash Payment
Tabby
Tamara


Customer
d
Customer-type
ISA
Loyal Customer
New Customer

3. Relational Model Design



4. Database Implementation and SQL Queries

```
Project                                                          1/6

CREATE TABLE Category (
    CategoryID NUMBER(9),
    CategoryName VARCHAR2(40) CONSTRAINT Category_Name_NN NOT N
    Apparel_Flag NUMBER(1) DEFAULT 0,
    Electronics_Flag NUMBER(1) DEFAULT 0,
    Makeup_Flag NUMBER(1) DEFAULT 0,
    Fragrance_Flag NUMBER(1) DEFAULT 0,
    Toys_Flag NUMBER(1) DEFAULT 0,
    Antiques_Flag NUMBER(1) DEFAULT 0,
    Furniture_Flag NUMBER(1) DEFAULT 0,
    CONSTRAINT PK_Category PRIMARY KEY (CategoryID),
    CONSTRAINT UQ_Category_Name UNIQUE (CategoryName),
    CONSTRAINT CHK_Apparel_Flag CHECK (Apparel_Flag IN (0, 1)),
    CONSTRAINT CHK_Electronics_Flag CHECK (Electronics_Flag IN
    CONSTRAINT CHK_Makeup_Flag CHECK (Makeup_Flag IN (0, 1)),
    CONSTRAINT CHK_Fragrance_Flag CHECK (Fragrance_Flag IN (0,
    CONSTRAINT CHK_Toys_Flag CHECK (Toys_Flag IN (0, 1)),
    CONSTRAINT CHK_Antiques_Flag CHECK (Antiques_Flag IN (0, 1)
    CONSTRAINT CHK_Furniture_Flag CHECK (Furniture_Flag IN (0,
);

CREATE TABLE Customer2(
  CustomerID2 NUMBER(9),
  FirstName2 VARCHAR2(30) CONSTRAINT Customer_FirstName_NN2 NOT
  LastName2 VARCHAR2(30) CONSTRAINT Customer_LastName_NN2  NOT
  Email2 VARCHAR2(80) CONSTRAINT Customer_Email_NN2 NOT NULL,
  PhoneNumber2 VARCHAR2(20),
  Address2 VARCHAR2(150),
  Country2 VARCHAR2(40),
  LoyaltyPointsBalance2 NUMBER DEFAULT 0,
  Customer_Type2 VARCHAR2(10) CONSTRAINT Customer_Type_NN2 NOT
  CONSTRAINT PK_Customer2 PRIMARY KEY (CustomerID2),
  CONSTRAINT UQ_Customer_Email2 UNIQUE (Email2),
  CONSTRAINT UQ_Customer_Phone2 UNIQUE (PhoneNumber2),
  CONSTRAINT CK_Customer_Points2 CHECK (LoyaltyPointsBalance2 >
  CONSTRAINT CK_Customer_Type2 CHECK (Customer_Type2 IN ('Loyal
);
```

1.1 / 6 2025.12.01 17:43:32

```
CREATE TABLE Product1 (
  ProductID NUMBER(9),
  ProductName VARCHAR2(80) CONSTRAINT Product_Name_NN NOT NULL,
  Description1 VARCHAR2(250),
  Price NUMBER(10,2) CONSTRAINT Product_Price_NN NOT NULL,
  StockQuantity NUMBER DEFAULT 0,
  CategoryID NUMBER(9) CONSTRAINT Product CategoryID NN NOT NUI
  CONSTRAINT PK_Product1 PRIMARY KEY (ProductID),
  CONSTRAINT CK_Product_Price CHECK (Price > 0),
  CONSTRAINT CK_Product_Stock CHECK (StockQuantity >= 0),
  CONSTRAINT FK_Product1_Categoryy FOREIGN KEY (CategoryID)
    REFERENCES Categoryy(CategoryyID)
);

CREATE TABLE Orders (
  OrderID NUMBER(9),
  OrderDate DATE CONSTRAINT Orders_Date_NN NOT NULL,
  CustomerID NUMBER(9) CONSTRAINT Orders_CustomerID_NN NOT NULI
  PaymentID NUMBER(9),
  OrderStatus VARCHAR2(20) CONSTRAINT Orders Status NN NOT NULI
  CONSTRAINT PK_Orders PRIMARY KEY (OrderID),
  CONSTRAINT CK_Orders_Status CHECK (OrderStatus IN ('PENDING',
  CONSTRAINT FK_Orders_Customer2 FOREIGN KEY (CustomerID)
    REFERENCES Customer2(CustomerID2)
);

CREATE TABLE Order_Detail (
  OrderDetailID NUMBER(9),
  OrderID NUMBER(9) CONSTRAINT OD_OrderID_NN NOT NULL,
  ProductID NUMBER(9) CONSTRAINT OD_ProductID_NN NOT NULL,
  Quantity NUMBER CONSTRAINT OD_Quantity_NN NOT NULL,
  Price NUMBER(10,2) CONSTRAINT OD_Price_NN NOT NULL,
  Subtotal NUMBER(12,2) CONSTRAINT OD_Subtotal_NN NOT NULL,
  CONSTRAINT PK_OrderDetail PRIMARY KEY (OrderDetailID),
  CONSTRAINT UQ_OD_Order_Product UNIQUE (OrderID, ProductID),
  CONSTRAINT CK_OD_Qty CHECK (Quantity > 0),
  CONSTRAINT CK_OD_Price CHECK (Price > 0),
```

```
  CONSTRAINT FK_OD_Order FOREIGN KEY (OrderID) REFERENCES Order
  CONSTRAINT FK_OD_Product1 FOREIGN KEY (ProductID) REFERENCES
);

CREATE TABLE Inventory (
  InventoryID NUMBER(9),
  ProductID NUMBER(9) CONSTRAINT Inventory ProductID NN NOT NUI
  QuantityAvailable NUMBER DEFAULT 0,
  ReorderLevel NUMBER DEFAULT 5,
  CONSTRAINT PK_InventoryID PRIMARY KEY (InventoryID),
  CONSTRAINT UQ_Inventory_ProductID UNIQUE (ProductID),
  CONSTRAINT CK_Inv_Qty CHECK (QuantityAvailable >= 0),
  CONSTRAINT CK_Inv_Reorder CHECK (ReorderLevel >= 0),
  CONSTRAINT FK_Inventory_Product1 FOREIGN KEY (ProductID)
    REFERENCES Product1(ProductID)
);

CREATE TABLE Payment (
    PaymentID NUMBER(9),
    PaymentDate DATE CONSTRAINT Payment_Date_NN NOT NULL,
    PaymentMethod VARCHAR2(50),
    Amount NUMBER(10,2) CONSTRAINT Payment_Amount_NN NOT NULL,
    OrderID NUMBER(9) CONSTRAINT Payment_OrderID_NN NOT NULL,
    PaymentStatus VARCHAR2(20) CONSTRAINT Payment Status NN NOT
    Payment_Type VARCHAR2(10) CONSTRAINT Payment_Type_NN NOT NU
    Card_Number VARCHAR2(16),
    Tabby_Tamara VARCHAR2(50),
    COD NUMBER(10,2),
    CONSTRAINT PK_Payment PRIMARY KEY (PaymentID),
    CONSTRAINT UQ_Payment_Order UNIQUE (OrderID),
    CONSTRAINT CK_Payment_Amount CHECK (Amount > 0),
    CONSTRAINT CK_Payment_Status CHECK (PaymentStatus IN ('PENI
    CONSTRAINT CK_Payment_Type CHECK (Payment_Type IN ('CARD','
    CONSTRAINT CK_Payment_Data CHECK (
        (Payment Type = 'CARD' AND Card Number IS NOT NULL AND
        (Payment_Type = 'DEFERRED' AND Card_Number IS NULL AND
        (Payment_Type = 'CASH' AND Card_Number IS NULL AND Tabb
```

```
    CONSTRAINT FK_Payment_Order FOREIGN KEY (OrderID) REFERENCE
);

ALTER TABLE Orders
ADD CONSTRAINT FK_Orders_PaymentID FOREIGN KEY (PaymentID) REFE

REATE TABLE Loyalty_Point (
  LoyaltyTransactionID NUMBER(9),
  Customer_ID NUMBER(9) CONSTRAINT LP_CustomerID_NN NOT NULL,
  PointsEarned NUMBER DEFAULT 0,
  PointsRedeemed NUMBER DEFAULT 0,
  PointsDate DATE CONSTRAINT LP_Date_NN NOT NULL,
  CONSTRAINT PK_LoyaltyPoint PRIMARY KEY (LoyaltyTransactionID)
  CONSTRAINT CK_LP_Earned CHECK (PointsEarned >= 0),
  CONSTRAINT CK_LP_Redeemed CHECK (PointsRedeemed >= 0),
  CONSTRAINT FK_LP_Customer2 FOREIGN KEY (Customer_ID)
    REFERENCES Customer2(CustomerID2)
);

-- Categoryy
INSERT INTO Categoryy VALUES (10, 'Apparel', 1, 0, 0, 0, 0, 0,
INSERT INTO Categoryy VALUES (20, 'Electronics', 0, 1, 0, 0, 0,
INSERT INTO Categoryy VALUES (30, 'Makeup', 0, 0, 1, 0, 0, 0, 0
INSERT INTO Categoryy VALUES (40, 'Fragrance', 0, 0, 0, 1, 0, 0
INSERT INTO Categoryy VALUES (50, 'Furniture', 0, 0, 0, 0, 0, 0
INSERT INTO Categoryy VALUES (60, 'Toys', 0, 0, 0, 0, 1, 0, 0);
-- Customer
INSERT INTO Customer2 VALUES (1,'Mayar','Ali','mayar@mail.com',
INSERT INTO Customer2 VALUES (2,'Sara','Hassan','sara@mail.com'
-- Product
INSERT INTO Product1 VALUES (1001, 'T-Shirt','Cotton t-shirt',
INSERT INTO Product1 VALUES (1002, 'Earbuds','Noise cancelling'
INSERT INTO Product1 VALUES (1003, 'Lip Gloss','Glossy finish',
INSERT INTO Product1 VALUES (1004, 'Jeans','Denim jeans', 150.0
INSERT INTO Product1 VALUES (1005, 'Smart Watch','Waterproof wa
INSERT INTO Product1 VALUES (1006, 'Vanilla Perfume','Strong sc
```

```
INSERT INTO Product1 VALUES (1008, 'Woody Oud','Rich oud scent'
-- Inventory
INSERT INTO Inventory VALUES (501, 1001, 50, 10);
INSERT INTO Inventory VALUES (502, 1002, 30, 5);
INSERT INTO Inventory VALUES (503, 1003, 80, 15);

-- Payment
INSERT INTO Payment VALUES (8001, TO DATE('25-oct-2025','DD-MON
INSERT INTO Payment VALUES (8002, TO_DATE('25-oct-2025','DD-MON
INSERT INTO Payment VALUES (8003, TO_DATE('26-oct-2025','DD-MON
-- Loyalty points
INSERT INTO Loyalty_Point VALUES (60001, 1, 20, 0, TO_DATE('25-
INSERT INTO Loyalty_Point VALUES (60002, 1, 10, 0, TO_DATE('25-
INSERT INTO Loyalty_Point VALUES (60003, 2, 0, 5, TO_DATE('26-c

SELECT *
FROM Customer2
ORDER BY CustomerID2 DESC;

SELECT Customer_ID,
SUM(PointsEarned) AS TotalPoints
FROM Loyalty_Point
GROUP BY Customer_ID
HAVING SUM(PointsEarned) > 20;

SELECT Payment_Type,
SUM(Amount) AS TotalPaid
FROM Payment
GROUP BY Payment_Type
ORDER BY TotalPaid ASC;

SELECT ProductID,ProductName, Price
FROM product1
WHERE Price <= 50;

SELECT
  P.ProductName,
```

```
FROM
  Product1 P
INNER JOIN
  Categoryy C ON P.CategoryID = C.CategoryyID
ORDER BY
  C.CategoryyName ASC;

SELECT
  C.FirstName2,
  O.OrderDate,
  P.PaymentStatus
FROM
  Orders O
INNER JOIN
  Customer2 C ON O.CustomerID = C.CustomerID2
INNER JOIN
  Payment P ON O.PaymentID = P.PaymentID
ORDER BY
  O.OrderDate DESC;

SELECT
  C.CategoryyName,
  COUNT(P.ProductID) AS NumberOfProducts
FROM
  Categoryy C
LEFT JOIN
  Product1 P ON C.CategoryyID = P.CategoryID
GROUP BY
  C.CategoryyName
ORDER BY
  NumberOfProducts DESC, C.CategoryyName ASC;
```

Queries output :

```sql
SELECT
  C.FirstName2,
  O.OrderDate,
  P.PaymentStatus
FROM
  Orders O
INNER JOIN
  Customer2 C ON O.CustomerID = C.CustomerID2
INNER JOIN
  Payment P ON O.PaymentID = P.PaymentID
ORDER BY
  O.OrderDate DESC;
```

100 % No issues found

Results Messages

	FirstName2	OrderDate	PaymentStatus
1	Mayar	2025-10-26 00:00:00.0000000	PENDING
2	Mayar	2025-10-25 00:00:00.0000000	PAID
3	Sara	2025-10-25 00:00:00.0000000	PAID

```sql
SELECT Customer_ID,
SUM(PointsEarned) AS TotalPoints
FROM Loyalty_Point
GROUP BY Customer_ID
HAVING SUM(PointsEarned) > 20;
```

100 % No issues found

Results Messages

	Customer_ID	TotalPoints
1	1	30

```sql
SELECT Payment_Type,
SUM(Amount) AS TotalPaid
FROM Payment
GROUP BY Payment_Type
ORDER BY TotalPaid ASC;
```

100 % No issues found

Results Messages

	Payment_Type	TotalPaid
1	DEFERRED	59.00
2	CARD	163.00
3	CASH	199.00

```sql
SELECT ProductID, ProductName, Price
FROM Product1
WHERE Price <= 50;
```

100 % No issues found

Results Messages

	ProductID	ProductName	Price
1	1003	Lip Gloss	45.00

```sql
SELECT
  C.CategoryName,
  COUNT(P.ProductID) AS NumberOfProducts
FROM
  Category C
LEFT JOIN
  Product1 P ON C.CategoryID = P.CategoryID
GROUP BY
  C.CategoryName
ORDER BY
  NumberOfProducts DESC, C.CategoryName ASC;
```

100 % No issues found

Results Messages

	CategoryName	NumberOfProducts
1	Fragrance	3
2	Apparel	2
3	Electronics	2
4	Makeup	1
5	Furniture	0
6	Toys	0

```sql
SELECT *
FROM Customer2
ORDER BY CustomerID2 DESC;
GO
```

100 % No issues found

Results Messages

	CustomerID2	FirstName2	LastName2	Email2	PhoneNumber2	Address2	Country2	LoyaltyPointsBalance2	Customer_Type2
1	2	Sara	Hassan	sara@mail.com	0500000002	Jeddah	Saudi Arabia	0	New
2	1	Mayar	Ali	mayar@mail.com	0500000001	Madinah	Saudi Arabia	120	Loyal

```sql
SELECT
  P.ProductName,
  C.CategoryName
FROM
  Product1 P
INNER JOIN
  Category C ON P.CategoryID = C.CategoryID
ORDER BY
  C.CategoryName ASC;
```

100 % No issues found

Results Messages

	ProductName	CategoryName
1	T-Shirt	Apparel
2	Jeans	Apparel
3	Earbuds	Electronics
4	Smart Watch	Electronics
5	Vanilla Perfume	Fragrance
6	Rose Scent	Fragrance
7	Woody Oud	Fragrance
8	Lip Gloss	Makeup

5.Conclusion

This project produced a complete database design for an Online Retail Store that reflects real E commerce operations. It helped us learn and practice the full database development process, especially EER modeling, mapping relationships into tables, applying primary and foreign keys for integrity, and writing SQL statements to create, populate, and query the database. Overall, our solution meets real world database requirements by organizing the store's data in a structured way and ensuring accuracy, consistency, and readiness for typical business operations such as order processing, payment tracking, inventory control, and loyalty point management.